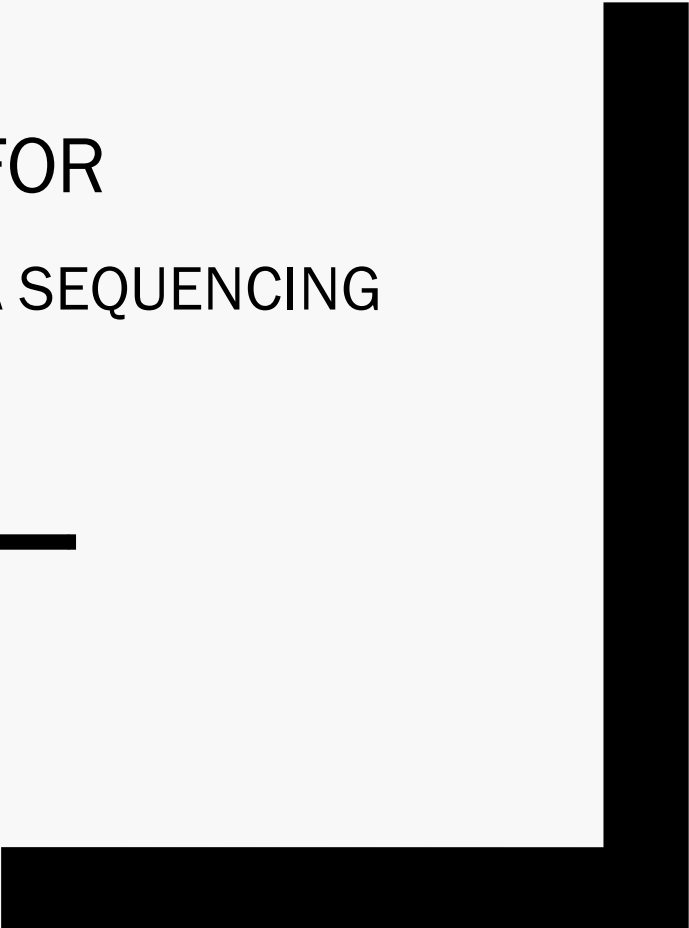




COMPUTATIONAL STRATEGY FOR CTNDA & MRD DETECTION FROM ONT PLASMA SEQUENCING

Jesús Urtasun-Elizari

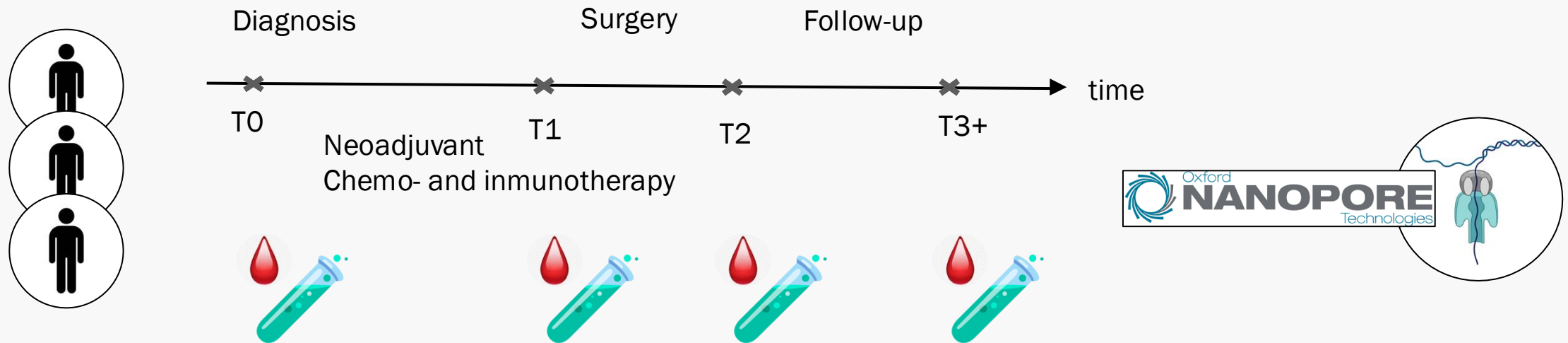
September 21st, 2026
MRC LMS Bioinformatics



Framing the problem

Experimental design & timepoints

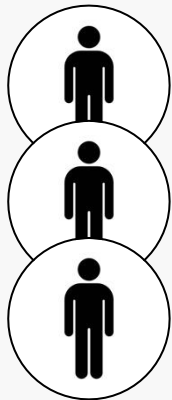
- Ovarian cancer patients, plasma cfDNA sampled at 4 timepoints



- ctDNA fraction highest at diagnosis, **drops after each intervention**

Framing the problem

The clinical questions & expected signal

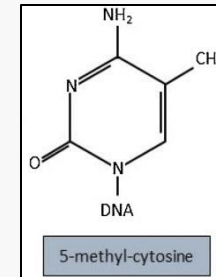
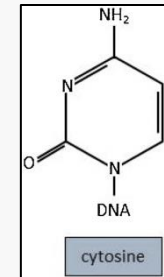


- Tumor and normal cfDNA mixed in unknown ratios per timepoint



- Signal we expect to detect:
 - *Differential 5mC between tumor and normal populations*

- Detection should scale with tumor fraction
 - *High at T0, vanishing toward T3+*

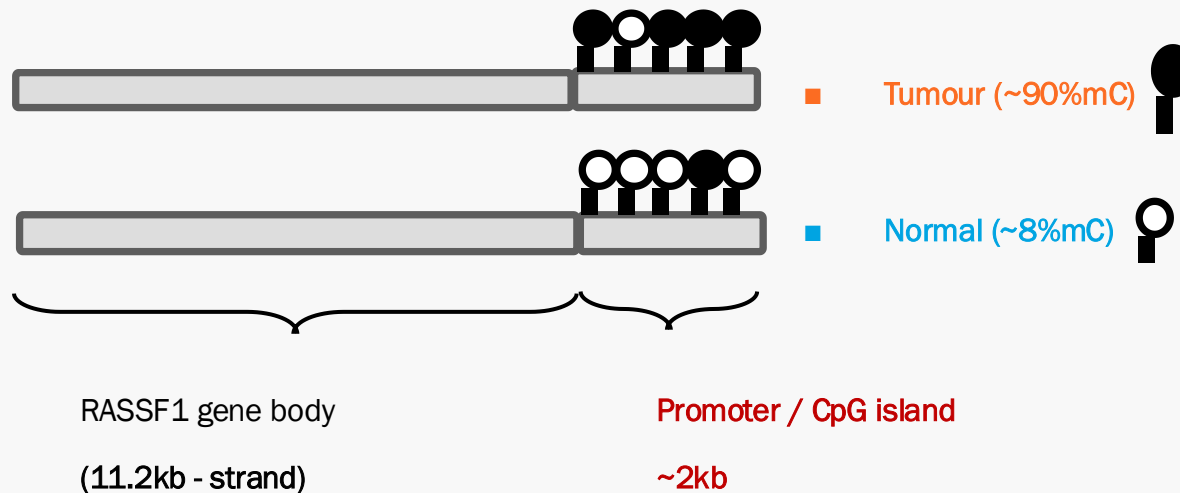


- Methylation vs variants vs fragmentomics (...)
- Can ONT sequencing detect ctDNA-associated methylation signal across TFG?
 - *MRD isn't a single measurable quantity, it's an operational threshold.*

Framing the problem

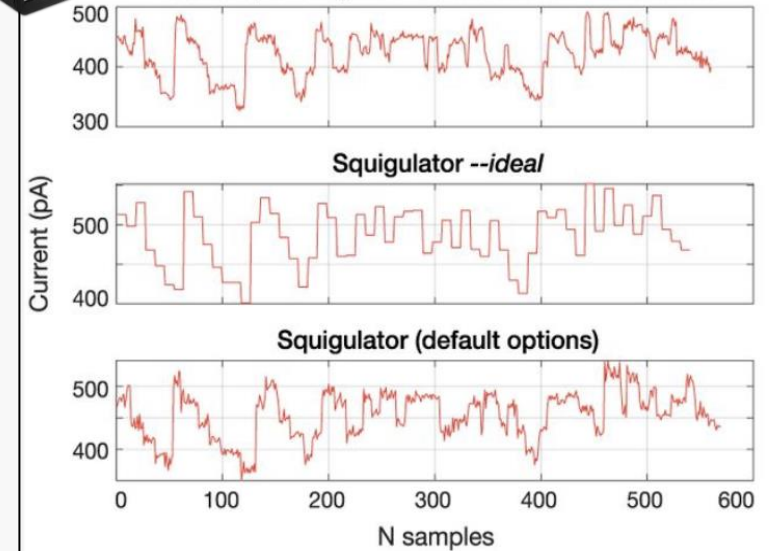
Simulating cfDNA & ctDNA toy data

- Raw ONT signal simulated with Squigulator,
 - From known ground-truth methylation frequencies
- Used the RASSF1A promoter as template locus
 - A known tumor suppressor commonly silenced by hypermethylation
 - Well-characterized as a methylation biomarker across cancer types



Experimental vs. simulated data

DNA sequencing on PromethION R9.4.1 flow-cell



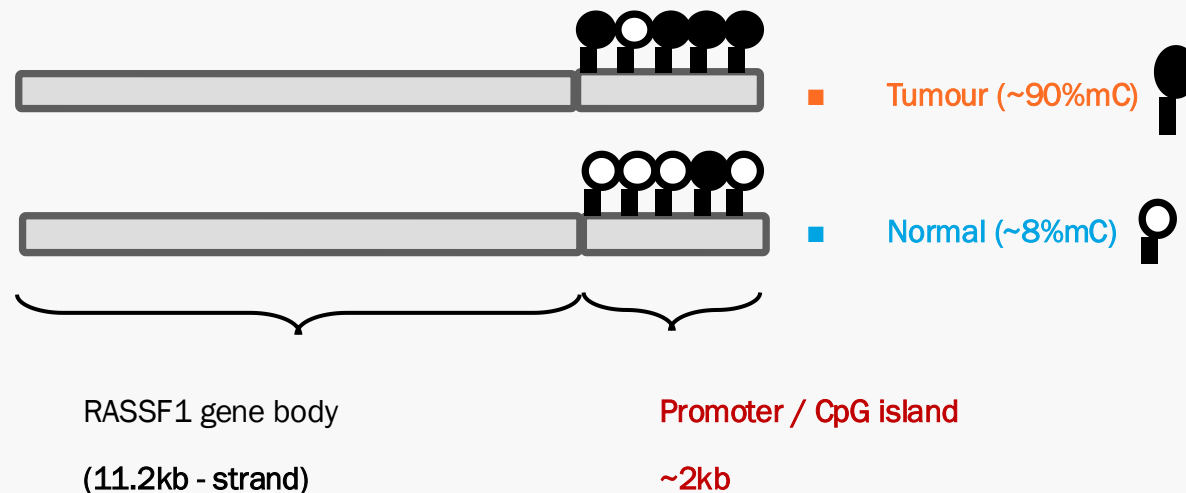
Hasindu Gamaarachchi et al., *Simulation of nanopore sequencing signal data with tunable parameters*,
Genome Research, <https://doi.org/10.1101/gr.278730.123>

A M Grawenda & E O'Neill, *Clinical utility of RASSF1A methylation in human malignancies*,
Br J Cancer 113, 372–381, <https://doi.org/10.1038/bjc.2015.221>

Framing the problem

Simulating cfDNA & ctDNA toy data

- Raw ONT signal simulated with Squigulator,
 - *From known ground-truth methylation frequencies*
- Used the RASSF1A promoter as template locus
 - *A known tumor suppressor commonly silenced by hypermethylation*
 - *Well-characterized as a methylation biomarker across cancer types*



- Assumptions:
 - Single locus rather than a multi-gene panel
 - Fixed per-population methylation level
 - Simulated rather than sequenced reads
 - Squigulator draws each call from fixed probabilities, rather than something measured from real calls

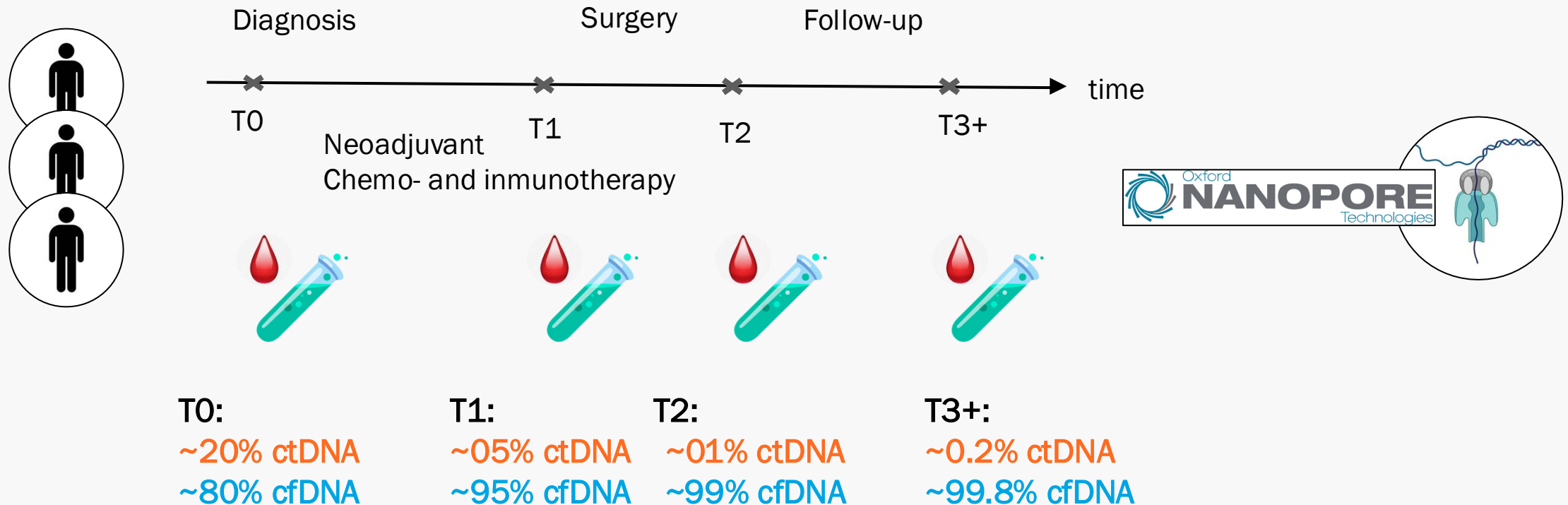
Hasindu Gamaarachchi et al.,
Simulation of nanopore sequencing signal data with tunable params.,
Genome Research, <https://doi.org/10.1101/gr.278730.123>

A M Grawenda & E O'Neill,
Clinical utility of RASSF1A methylation in human malignancies,
Br J Cancer 113, 372–381, <https://doi.org/10.1038/bjc.2015.221>

Framing the problem

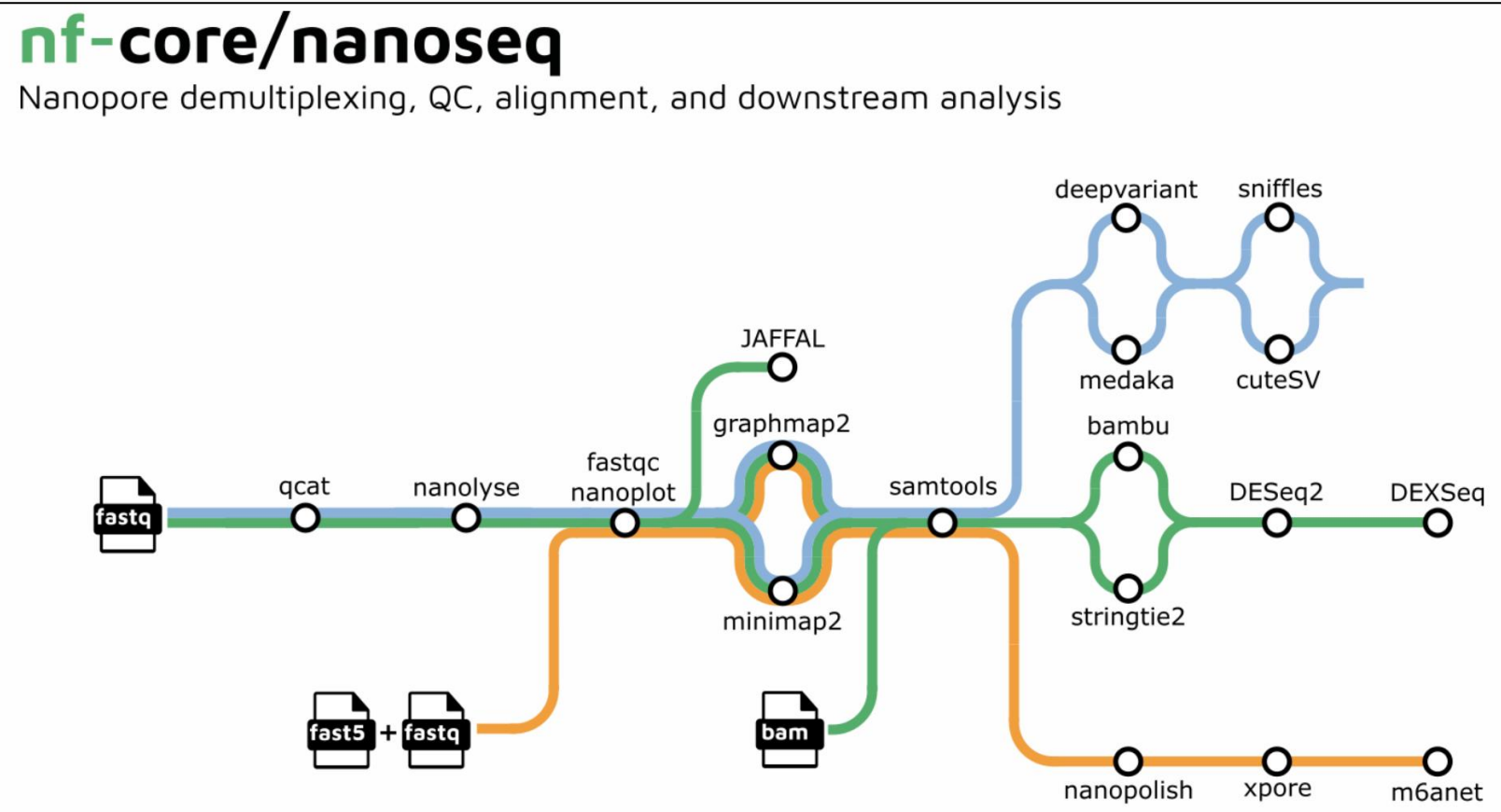
Simulating cfDNA & ctDNA toy data

- Ovarian cancer patients, plasma cfDNA sampled at 4 timepoints



Pipeline design

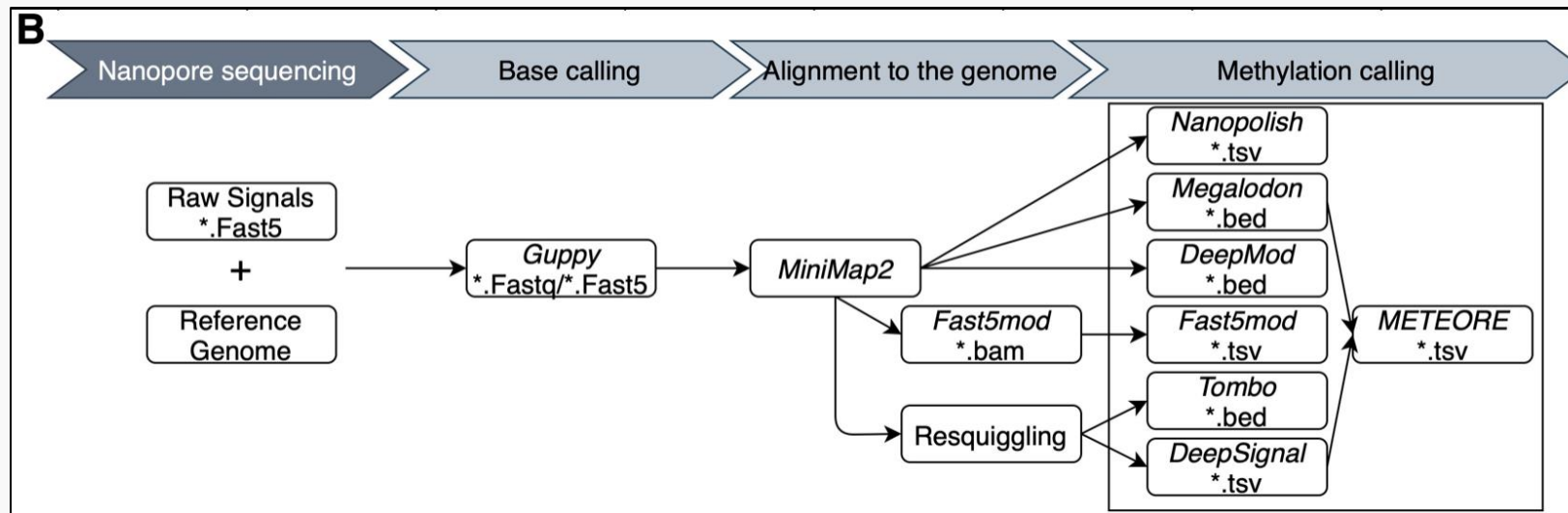
Standard ONT & specifics for ctDNA detection



nf-core/nanoseq — community-maintained Nextflow pipeline for nanopore demultiplexing, QC, alignment, and methylation calling: github.com/nf-core/nanoseq

Pipeline design

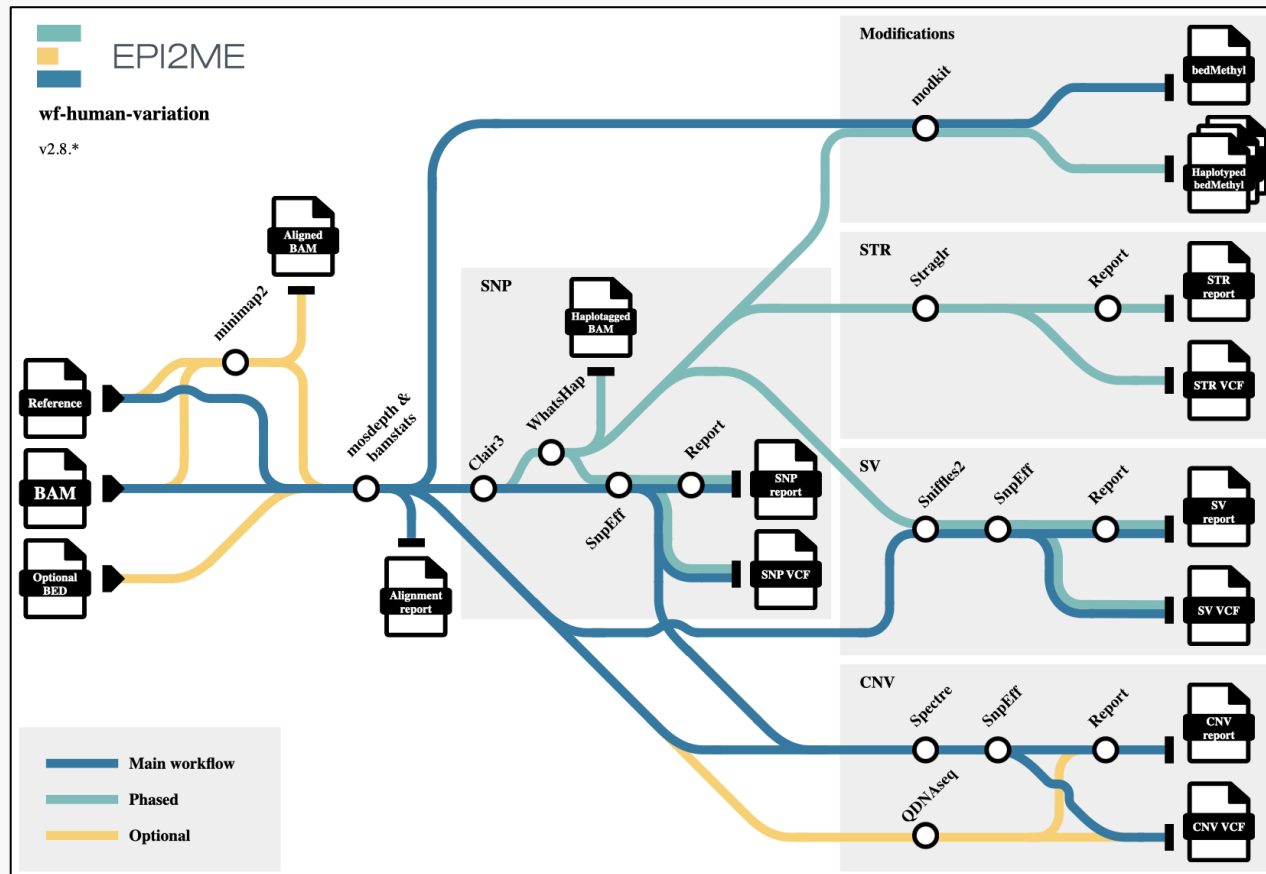
Standard ONT & specifics for ctDNA detection



EPI2ME Labs wf-human-variation — Oxford Nanopore's own official workflow, includes basecalling, alignment, variant calling and 5mC/5hmC methylation modules: github.com/epi2me-labs/wf-human-variation

Pipeline design

Standard ONT & specifics for ctDNA detection



NANOME — published consensus nanopore DNA-methylation-detection pipeline, combining multiple modbase-calling tools for higher-confidence calls: github.com/LabShengLi/nanome

Pipeline design

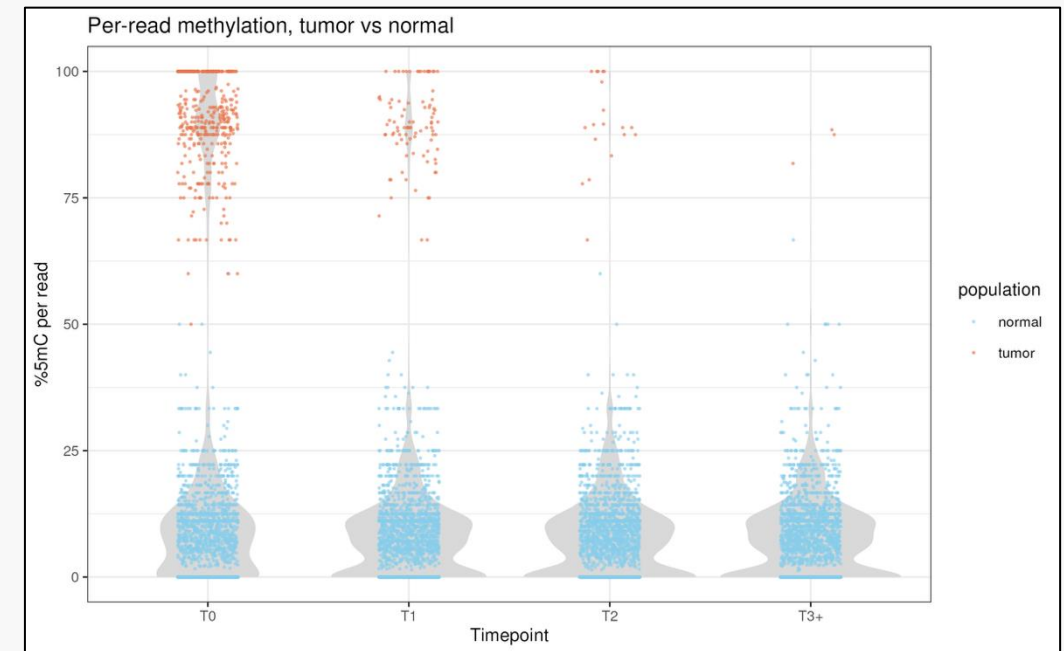
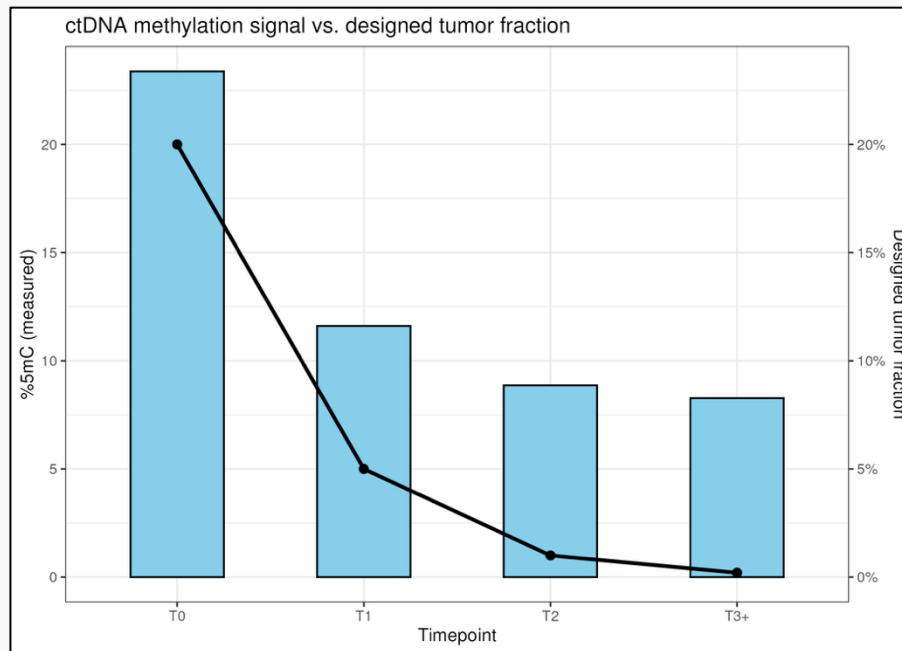


Standard ONT & specifics for ctDNA detection

- Pipeline: squigulator → basecall/tag → minimap2 align → samtools sort/index → modkit pileup (or custom decode)
- Tricky point 1: minimap2's -y doesn't flip MM/ML tags on strand-reversed reads — needed a manual tag-regeneration step
- Tricky point 2: modkit panicked on our data with an unresolved `InvalidDnaBase` error — bypassed with a custom Python MM/ML decoder

Results

Signal Degrades Under Realism



- Naive: %5mC signal tracks designed tumor fraction closely and proportionally
- Realistic: signal is compressed — shorter tumor reads carry fewer CpG sites per read, diluting the aggregate methylation estimate

Limitations of simulated ONT data

Three realism edits

- Low tumor fraction: ctDNA can be well under 1% VAF, especially post-treatment
- Nanopore's lower per-read accuracy raises the noise floor for rare-signal calling
- Fragment biology matters: cfDNA length and end-motifs carry tumor-associated information independent of methylation
- Our first pipeline worked, but tested none of these — so we rebuilt the simulation to include them

Limitations of simulated ONT data



Three realism edits

- Per-read modification noise: widened/overlapping methylation-probability ranges, mimicking nanopore's per-read basecalling error
- End-motif bias: population-specific 5' fragment motifs, mimicking non-random nuclease cleavage during cfDNA release
- Fragment-length split: tumor fragments simulated shorter than normal (220bp vs 400bp), mimicking tumor cfDNA's more irregular fragmentation

Why naive simulation isn't enough

Three realism edits

- Fragment length, tumor cfDNA shorter, more irregularly fragmented

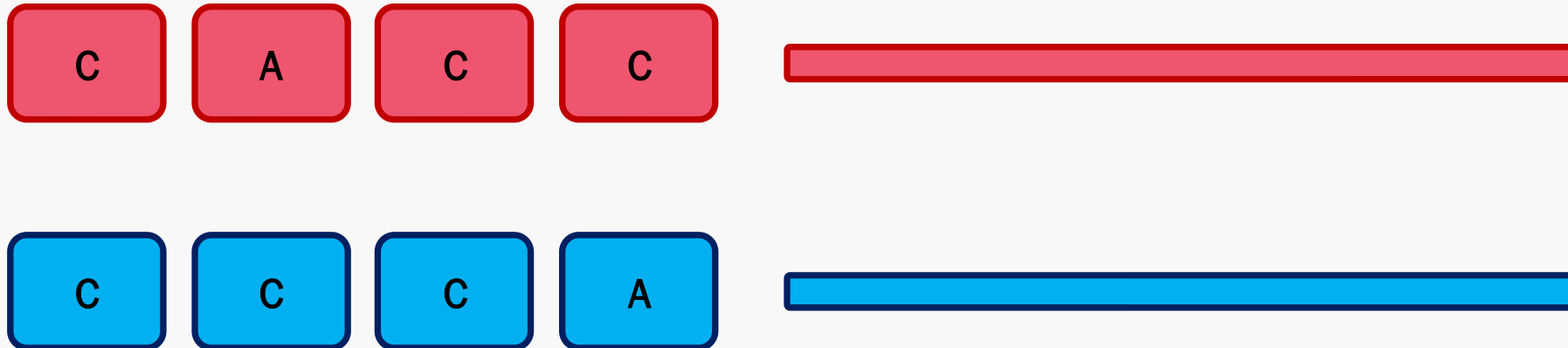
Tumour ~220bp

Normal ~400bp

Why naive simulation isn't enough

Three realism edits

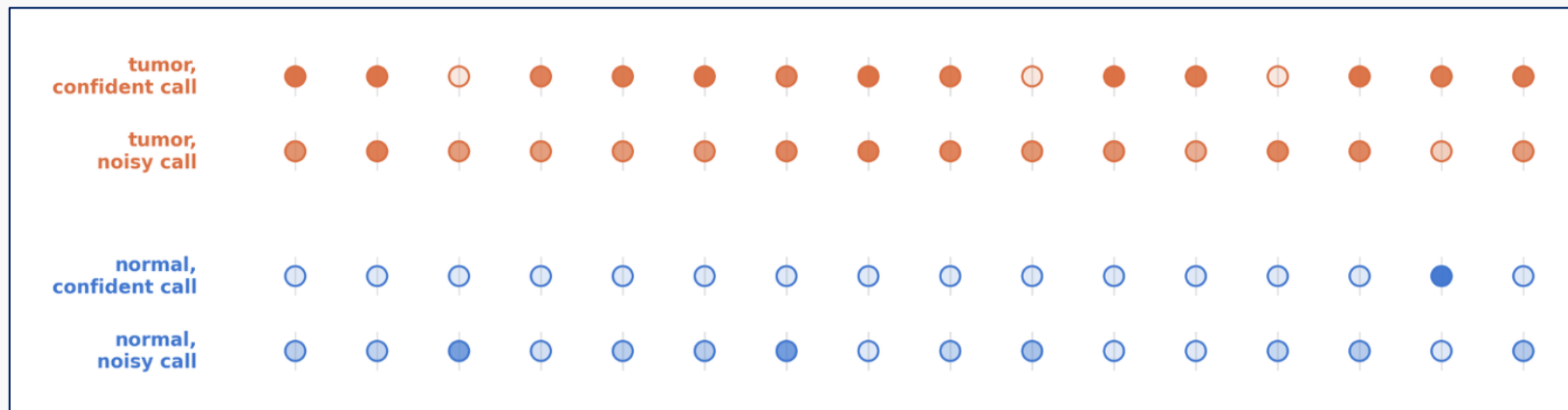
- End-motif bias: population-specific 5' cleavage signature (4-mer)



Why naive simulation isn't enough

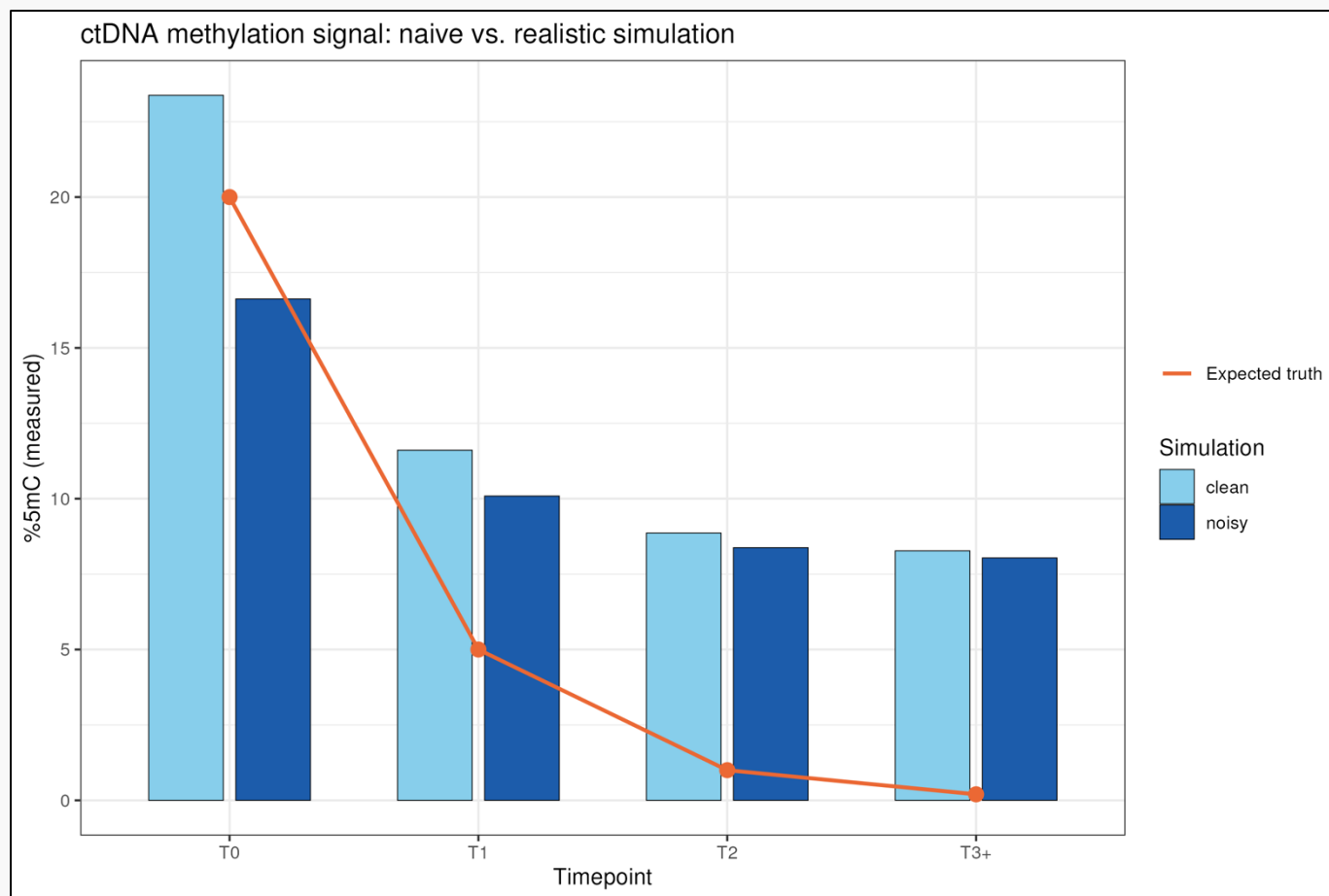
Three realism edits

- Per-read modification noise: same true state, but wider/overlapping called probability



Results

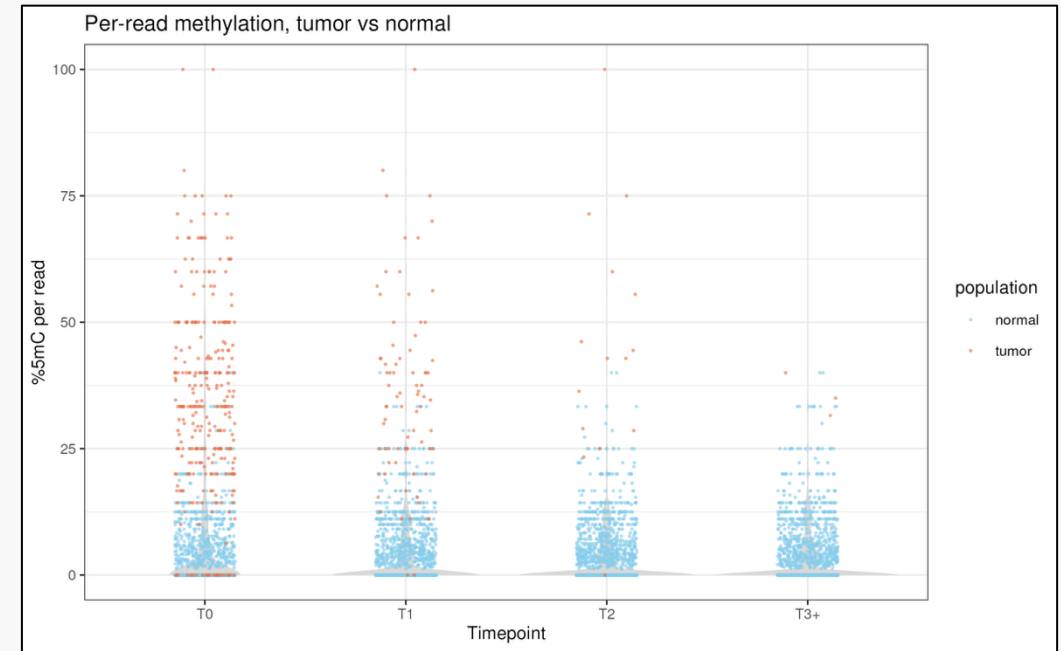
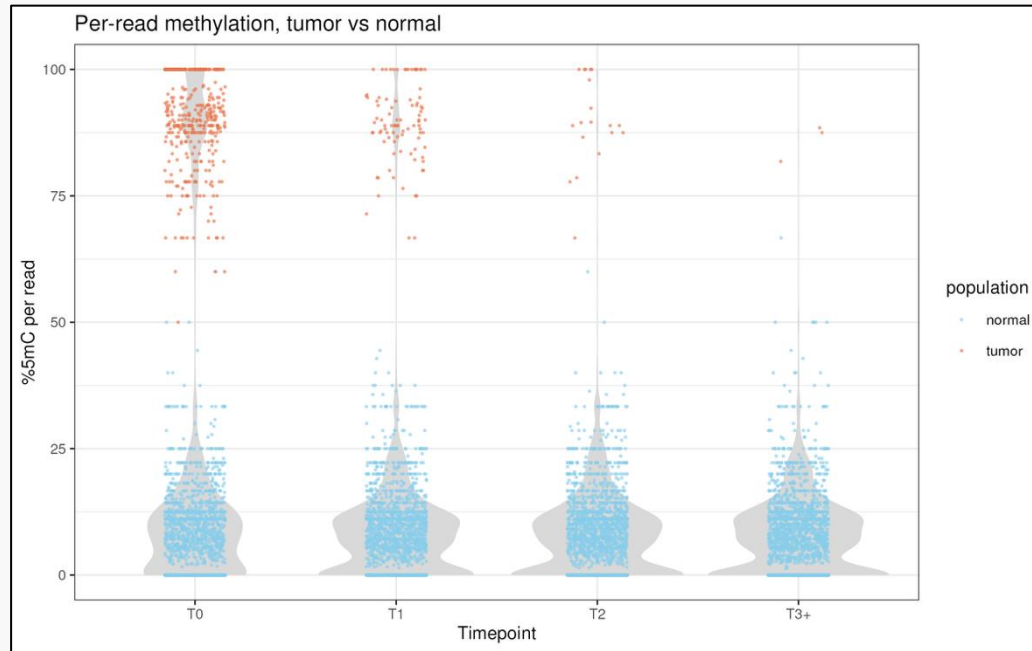
Signal Degrades Under Realism



- Naive: %5mC signal tracks designed tumor fraction closely and proportionally
- Realistic: signal is compressed — shorter tumor reads carry fewer CpG sites per read, diluting the aggregate methylation estimate

Results

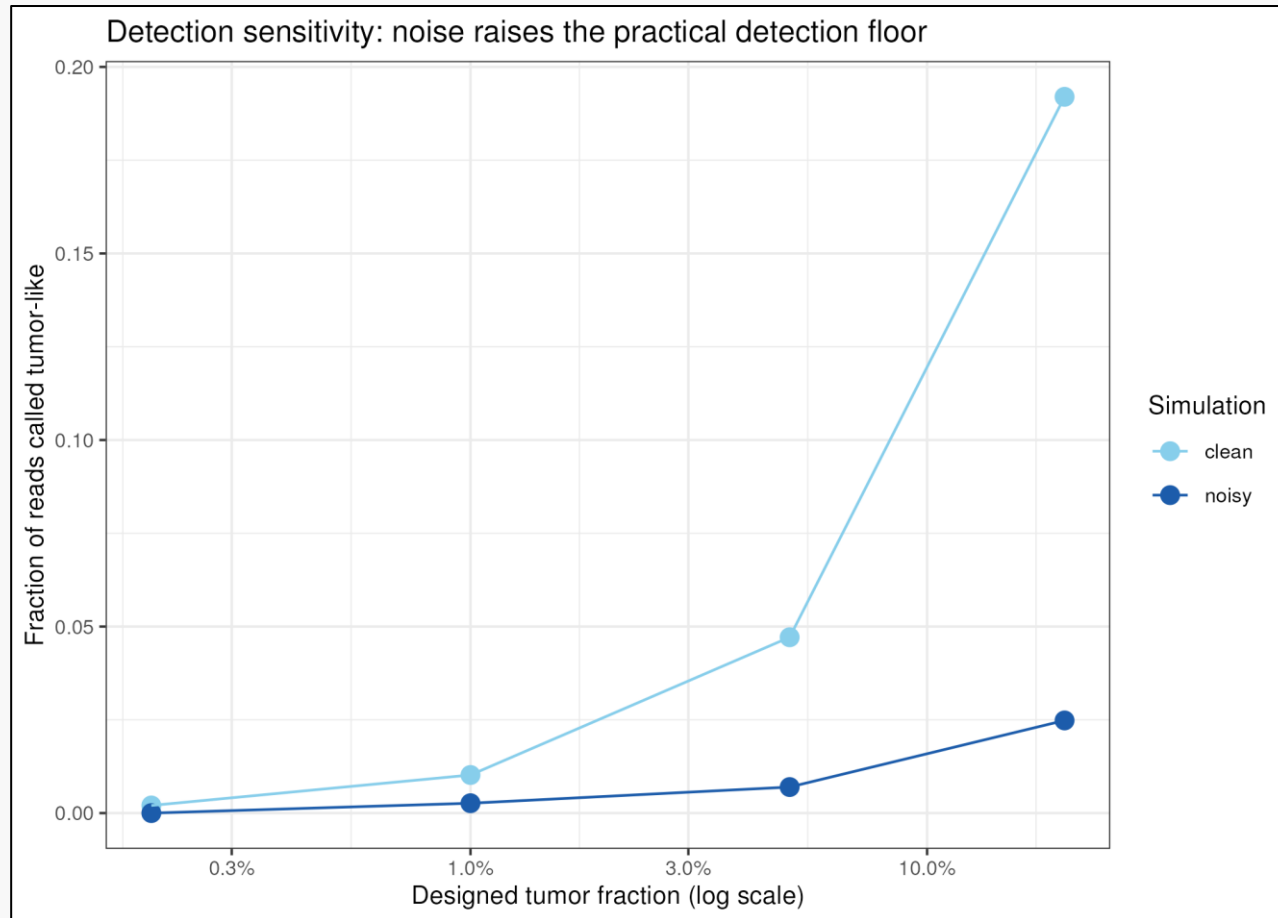
Signal Degrades Under Realism



- Naive: %5mC signal tracks designed tumor fraction closely and proportionally
- Realistic: signal is compressed — shorter tumor reads carry fewer CpG sites per read, diluting the aggregate methylation estimate

Results

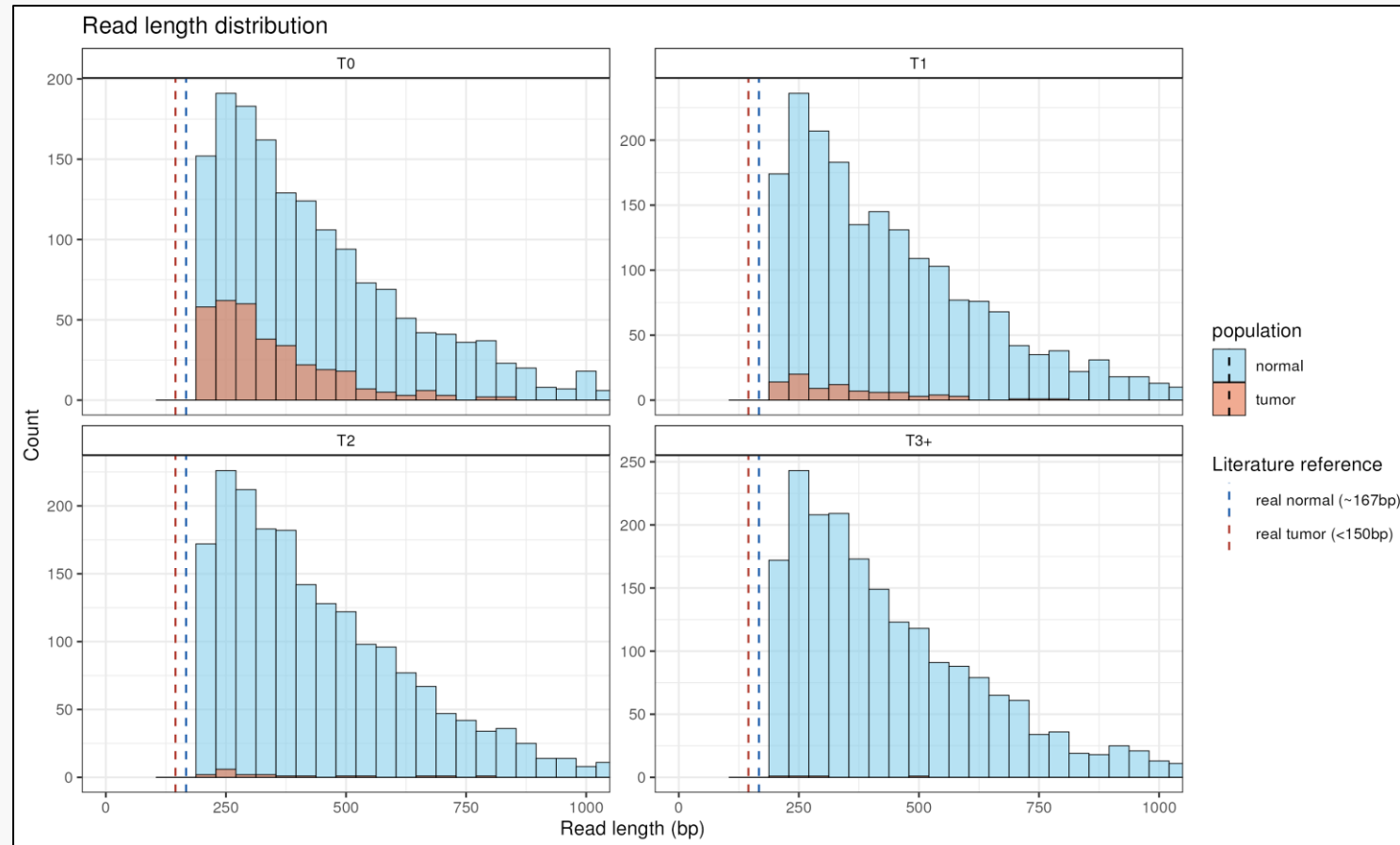
Signal Degrades Under Realism



- Naive: %5mC signal tracks designed tumor fraction closely and proportionally
- Realistic: signal is compressed — shorter tumor reads carry fewer CpG sites per read, diluting the aggregate methylation estimate

Results

Limitations We Identified, Not Missed



- Naive: %5mC signal tracks designed tumor fraction closely and proportionally
- Realistic: signal is compressed — shorter tumor reads carry fewer CpG sites per read, diluting the aggregate methylation estimate

Results



Limitations We Identified, Not Missed

- Real MRD detection must survive exactly what broke our simulation: low signal, per-read noise, short fragments
- Argues for multi-locus, multi-signal aggregation over single-locus, per-read calling
- Next step if this were a real assay: expand to a larger methylation panel, increase depth, and incorporate fragment-length/end-motif as an independent, orthogonal classifier alongside methylation

Questions?



Signal choice

From variants to methylation and fragmentomics

- Somatic variants (SNP / Indels)
 - *require tumour-informed panel for sensitivity*
 - *ONT per-read error raises the noise floor of rare alleles*
 - *Cofounded by clonal hematopoiesis (CHIP)*
- Methylation (modkit / DMRs – 5mC & 5hmC)
 - *Native modified-base calling, no extra chemistry*
 - *Strong, reproducible tumour-vs-normal DMRs even tumour-naïve*
 - *Directly reuses an existing dorado-modkit-DMR pipeline*
- Fragmentomics (size, end-motif, nucleosome footprint)
 - *Genome-wide signal, robust at low coverage*
 - *Complementary, largely orthogonal to sequence & methylation*
 - *Well suited to long, native ONT fragments*

Caveats



Cofounders to control prior to trusting signal

- Clonal hematopoiesis (CHIP):
 - *Age related clonal variants in white blood cells mimic tumour variants – needs matched WBC sequencing to filter.*
- Low input / degraded DNA:
 - *Post-surgery and follow-up samples may have very low cfDNA yield, limiting depth and sensitivity. LoD must be sample specific.*
- Cross-timepoint batch effects:
 - *T0 -> T3+ collected over months; extraction batch, library prep, and flow-cell run must be tracked as covariates.*
- Pannel / assay drift:
 - *If a tumour-informed panel is designed from T1/T2 tissue, ensure it is still applicable to earlier/later timepoints and relapse clones.*

Framing the problem



Simulating cfDNA & ctDNA toy data

- Raw ONT signal simulated with Squigulator, from known ground-truth methylation frequencies
- Used the RASSF1A promoter as template locus — a tumor suppressor commonly silenced by hypermethylation across cancer types, well-characterized as a methylation biomarker
- Assumptions
 - *Single locus: we only simulated the RASSF1A promoter (~2kb) rather than a multi-gene panel — a real assay would need many loci to reach clinical sensitivity.*
 - *Fixed per-population methylation level: every CpG in tumor-derived reads has a 90% methylation probability, and every CpG in normal-derived reads has 8% — flat across the whole locus and constant across all four timepoints; only the mixture ratio of these two populations changes over time.*
 - *Simulated rather than sequenced reads: these frequencies are simulator inputs (squigulator draws each call from these fixed probabilities), not something measured from real plasma — so the underlying truth is by construction, not biology-observed.*

Hasindu Gamaarachchi et al., Simulation of nanopore sequencing signal data with tunable parameters, Genome Research, <https://doi.org/10.1101/gr.278730.123>

A M Grawenda & E O'Neill, Clinical utility of RASSF1A methylation in human malignancies, Br J Cancer 113, 372–381, <https://doi.org/10.1038/bjc.2015.221>

Predict response to neoadjuvant immunotherapy



Realistic

- **Per-read modification noise** — widened/overlapping methylation-probability ranges per call, mimicking nanopore's per-read basecalling error rate (lower single-read accuracy than short-read), which raises the noise floor and produces ambiguous "filtered" calls near the 5mC/unmodified threshold rather than clean binary reads.
- **End-motif bias** — biased the 5' end of each fragment toward population-specific 4-mer motifs (e.g. CCCA-family for normal), mimicking non-random nuclease cleavage during apoptotic chromatin fragmentation, a methylation-independent signal known to carry tumor-vs-normal information in real cfDNA.
- **Fragment-length differentiation** — simulated tumor reads shorter than normal reads (220bp vs 400bp), reflecting that tumor-derived cfDNA is more fragmented/"jagged" from irregular nuclease digestion, versus normal cfDNA's tighter nucleosome-protected length distribution.

Predict response to neoadjuvant immunotherapy

F1

- **Yesterday (clean):** T0 bar $\approx 23.4\%$, dropping to 11.6/8.9/8.2%, tracking the designed fraction line fairly proportionally at the top end.
- **Today (3 realism edits):** T0 bar is now lower ($\approx 16.7\%$ vs $\approx 23.4\%$) and the whole curve is compressed — T1/T2/T3+ (10.1/8.4/8.1%) sit much closer together than yesterday's spread.
- **Interpretation:** the compression is expected and mostly attributable to the fragment-length change, not noise/motif — shorter tumor reads (220bp) carry fewer CpG sites per read than the old uniform-length reads, so tumor reads contribute proportionally less signal per read at high tumor fraction (T0), pulling the whole curve down and flattening the gradient. This is a realistic effect: real ctDNA fragments are short and CpG-sparse per read, which is exactly the kind of sensitivity loss a real assay would have to fight.
- Ready for Fig 2 (per-read violin) whenever you want to move on.

Predict response to neoadjuvant immunotherapy



F2

- **Yesterday (clean):** Tight, well-separated clusters — tumor reads sit almost entirely in the 75–95% band, normal reads almost entirely below 25%. Very little overlap between populations.
- **Today (3 realism edits):** Much messier — tumor points are now scattered across the full 0–100% range (down to ~0 at higher timepoints) and normal points reach further upward too. The clean separation between the two clouds is largely gone; a classifier drawing a single threshold would misclassify far more reads.
- **Interpretation:** this is exactly the expected effect of the widened per-read noise range plus the shorter tumor fragments — fewer CpG calls per tumor read (due to shorter length) means each read's %5mC estimate is noisier and more likely to land far from its true population mean, especially at low coverage (T2/T3+, where only ~4–20 tumor reads exist total, so a couple of noisy reads visibly disrupt the pattern). This is a much more honest depiction of what per-read nanopore classification would actually look like at low tumor fraction — real MRD calling can't rely on a clean per-read threshold and needs aggregation across many reads/positions.

Predict response to neoadjuvant immunotherapy

F3

- **Real values worth citing** (well-established in cfDNA fragmentomics literature, e.g. Underhill et al. 2016, Mouliere et al. 2018 *Sci Transl Med*): normal cfDNA is dominated by a mononucleosomal peak at **~167bp** (147bp nucleosome core + ~20bp linker); tumor-derived cfDNA is shifted shorter, enriched in **sub-nucleosomal fragments ~90–150bp**, well below what a healthy-cell apoptotic ladder produces.
- This turns the earlier problem into a legitimate finding for the interview: *"we identified that squigulator enforces a 200bp minimum read length, which prevents simulating true sub-nucleosomal ctDNA fragments (<150bp) — a known real biological signal we could describe but not fully reproduce with this tool."* That's a stronger answer than pretending the separation worked cleanly.
- labs(subtitle = "Our simulated means (220/400bp) are shifted right of biology, \nfloored by squigulator's 200bp minimum read length")

Predict response to neoadjuvant immunotherapy



F4

- **Yesterday (clean):** Smooth, well-separated bands — T0 sits clearly highest (~20–26%), T1 clearly second (~11–15%), and T2/T3+ overlap at the bottom (~5–11%) but the overall ordering is easy to read at a glance.
- **Today (3 realism edits, filtered):** Much lower ceiling (~6–12% max vs ~26%) and the bands are far noisier and closer together — T1/T2/T3+ now interleave and cross each other repeatedly rather than sitting in clean separate bands. T0 remains visibly above the rest, but the T1 vs T2 vs T3+ distinction that was clean yesterday is now largely lost in noise.
- **Interpretation:** consistent with everything else we've seen — shorter tumor fragments + per-read noise both reduce the number of confidently-called methylated CpGs contributing to each position's average, so the signal-to-noise ratio at the per-position level degrades substantially at low tumor fraction. This is a legitimate and important finding for the interview: it directly demonstrates *why* real ctDNA/MRD assays at T2/T3+-like tumor fractions can't rely on single-locus per-CpG resolution and need to aggregate across the whole region (or many loci) to get a usable signal — exactly the point Fig 1 (aggregate %5mC) is more robust to than Fig 4 (per-position) at low tumor fraction.

Predict response to neoadjuvant immunotherapy



F5

- **Yesterday (clean):** Smooth, monotonic build-up from 0 to a plateau (~750–950x) in the outer thirds of the locus, tapering at the far end — typical of uniform-length reads all spanning most of the amplicon, so coverage accumulates predictably as reads start extending across it.
- **Today (3 realism edits):** Dramatically sawtoothed — T0 and T3+ show repeated sharp drops to near-0 at multiple positions (e.g. ~700, 800, 900, 1100–1200, 1350, 1650, 1750bp) before spiking back up, while T1/T2 stay comparatively smoother, ramping to a lower plateau (~500–580x vs ~750–950x yesterday).
- **Interpretation:** this is the same fragment-length effect showing up again, but now visible in coverage rather than in methylation. Because reads are now much shorter (220/400bp) relative to the 2001bp locus, individual reads only cover a portion of it — so at any given position, coverage depends on how many of the (few, for tumor) short fragments happen to span that exact spot. With few tumor reads (T0 has only 400 vs T3+'s 4), a handful of fragment boundaries lining up creates visible coverage gaps; this is amplified further at T0's low NTUM because it's the population with the fewest, shortest fragments contributing at any position. This is a genuine artifact of simulating short fragments against a fixed-length locus with a small read pool — real WGS/targeted panels average this out with far higher depth, but it's a fair thing to flag as a limitation of simulating a single small locus rather than a full genomic region.

Predict response to neoadjuvant immunotherapy



F7

- **Yesterday (clean):** Sensitivity tracks tumor fraction fairly proportionally across the range — $T0 \approx 0.19$, $T1 \approx 0.047$, $T2 \approx 0.010$, $T3+ \approx 0.002$ — a roughly linear relationship on the log-log axes, detection never fully vanishes even at the lowest fraction.
- **Today (3 realism edits):** Ceiling drops sharply ($T0 \approx 0.025$, ~8x lower than yesterday), and **T3+ sensitivity goes to essentially zero** — at the lowest, most clinically relevant tumor fraction (designed 0.2%), the combination of noise + shorter/lower-signal tumor fragments pushes detection below the practical floor entirely.
- **Interpretation — this is the most important figure of the three-way comparison.** It's a direct, quantitative demonstration of the exact "hard constraint" you flagged earlier from the interview brief: *low tumor fraction + nanopore's per-read noise floor together can erase detectability at the timepoint that matters most for MRD (residual disease after treatment)*. This is a much stronger talking point than the clean version, which never showed real failure — real MRD detection has to solve exactly this problem (via deeper coverage, larger panels, or aggregating many loci), and this figure shows you've reproduced the failure mode rather than just the easy, high-tumor-fraction case.
- This is a strong figure to lead with in the technical discussion — it turns your "3 realism edits" into a concrete negative result that mirrors the real clinical difficulty, rather than just noisier versions of the same story.